

Wing morphometrics of Oregon *Melissodes* bees

Introduction

Oregon has over 600 species of native bees, spanning 55 genera, including bumblebees, leafcutter bees, mason bees, and mining bees (Cox et al. 2023). Native bees play crucial roles as pollinators in their ecosystems and in agriculture. *Melissodes*, common name “longhorn bees,” are members of the family Apidae, and there are up to 20 species native to Oregon. Members of the *Melissodes* genus are pollinators of sunflowers and related species (LaBerge 1961, Mallinger et al. 2018, Roch 2023).

Members of this genus have diverged relatively recently in evolutionary history. Species within the subgenus *Eumelissodes* diverged around 3 million years ago (Wright 2018). The current study focuses on two species, *M. agilis* and *M. trinodis* in the subgenus *Eumelissodes*. Due to morphological similarities, members of those species are near impossible to distinguish with the naked eye. The males of both species have distinctive yellow clypei, one yellow spot at the base of each mandible, and yellow labrums (LaBerge 1961). They have different ranges, with *M. agilis* being found throughout the continental United States, in southern Canada, and in northern Mexico, and *M. trinodis* being found in a similar range but not in the far western United States (Ascher and Pickering 2020). Roch (2023) attempted to distinguish between them using both phylogenetics and morphological analysis. Using just specimens that keyed to *M. agilis* or *M. trinodis*, Roch found 3-5 clades based on sequence data, suggesting that there may actually be 3-5 species in this cryptic species complex.

Phylogenetic research often uses the mitochondrial gene cytochrome oxidase 1 (COX1) as a DNA “barcode” to distinguish between species (Hebert 2003).

Unfortunately, COX1 can struggle with species level identification, especially when the species have diverged relatively recently. The COX1 reference sequences for *M. agilis* and *M. trinodis* published by Wright et al. (2020) differed by only three base pairs, which means other mechanisms of identification are needed to confidently differentiate between members of these two species.

Wing morphometrics is a tool for differentiating insect taxa developed in 1998 (Bookstein 1998). Insect wings have distinctive vein patterns that can be used to distinguish between species and sex across all of Insecta. Members of Hymenoptera have relatively simple venation patterns when compared with other orders of insects (Salcedo et al. 2019). The simple venation patterns mean that the difference between species can be so slight that they are not visible with the naked eye, which is where computerized analysis becomes important. Wing morphometrics is performed by landmarking each vein intersection on the chosen wing, minimizing the distances between each landmark using Procrustes analysis, and then performing PCA on the standardized landmarks. (Masaquiza et al. 2023, Roch 2023). For the subfamily Apinae there are 19 defined landmarks on the forewing at either vein intersections or interior apexes of major wing cells (Masaquiza et al. 2023, Roch 2023, Francoy et al. 2008). (Figure)

In bees alone, wing morphometrics have been used to identify gender and species of native honey bees in Thailand (Atsalek et al. 2010), distinguish between genders of five genera of stingless bees (Francoy et al. 2009), and identify African honeybees (Francoy et al. 2008). When a combined genetic and wing morphometric approach was

performed on specimens that keyed to *M. agilis* and *M. trinodis*, Roch (2023) found several distinct clades. Some specimens initially categorized as either *M. agilis* or *M. trinodis* formed a complex with Wright's (2020) reference sequences for *M. pallidisignatus*, *M. appressa*, and *M. interrupta*, some formed a complex with one another and no reference sequence, and most fell into a clade containing both *M. agilis* and *M. trinodis* reference sequences. The clade with *M. agilis* and *M. trinodis* had three subclades, one of which contained the *M. agilis* reference sequence, one of which contained no reference sequences, and one of which contained the *M. trinodis* reference sequence. He also found there was a strong sexual dimorphism in forewing shape across clades, which is not surprising given the sexual dimorphism seen in *Melissodes* bees (LaBerge 1956, 1961). He also found that forewing shape was correlated with genetic clade and, in some clades, could be used to sort specimens into them. Continued research into the wing shape of *M. agilis* and *M. trinodis* specimens can give further insight into these trends and potentially further strengthen the wing/clade relationship found in Roch (2023).

The current project hopes to explore the diversity within specimens from Oregon, particularly those that appear to be *M. agilis* or similar, and to determine whether the clades found by Roch are present here. *M. agilis* is found in Oregon and *M. trinodis* is not, but there are many specimens that are morphologically similar to *M. agilis*. We used wing morphometrics and phylogenetic analysis of COX1 sequence data to determine how many clades are present in Oregon, whether any of them are true *M. agilis*, and whether they align with the clades found by Roch (2023).

Methods

Specimen Collection

All specimens were collected near sunflowers around Oregon between 2017 and 2025. Specimens collected prior to 2025 were collected by contributors to the Oregon Bee Atlas, and the Oregon Bee Atlas taxonomist identified them to their genus. Specimens collected in 2025 were collected by Johnson or Lindh at Willamette University. All specimens were sorted to morphogroup by Lindh. Wings were collected from mainly *Eumelissodes* specimens, with an emphasis on *M. agilis* specimens or similar, building on the collection of wing photos started by Apio and Strobel (2025). Total number of subgenera amounted to four: *Melissodes* (Mel.), *Eumelissodes* (Eu.), *Heliomelissodes* (Helio.), and *Callimelissodes* (Calli.). [Table S1](#) contains details for specimen collectors, collection dates, and locations.

Wing Photography

The right forewing was removed from each specimen, placed on a glass microscope slide, and held in place by a coverslip secured with drops of clear nail polish or xylene at each corner. Each wing was then photographed with backlighting with an Olympus stereomicroscope.

Landmarking

If wings were placed upside down, the photos were reflected across the y-axis such that the shoulder was on the left side, and the end of the wing was on the right. Wings were then landmarked at the 19 vein intersections used in Roch (2023) using the Stereomorph package in R (Olsen and Westneat 2015). Wings with any missing landmarks were left out of analysis.

DNA Collection

For bees collected in 2025, DNA extraction was performed according to the Willamette University Biology 245W Protocol using a Qiagen DNeasy kit (Lindh and Smith 2024). Two right legs were pulled from each specimen and shattered with a plastic pestle after dipping the tube in liquid nitrogen. Buffer ATL and Proteinase K were added, and tubes were then vortexed and incubated in a heat block for 24 hours. After the incubation period, Buffer AL and ethanol were added to each tube and immediately vortexed. The mixture was then added to a spin column and centrifuged. Buffers AW1 and AW2 were added, centrifuging after the addition of each one. Buffer AE was added and centrifuged one final time. Samples were stored in a refrigerator set at 4° C until use.

Polymerase Chain Reaction

PCR was performed according to the Willamette University Biology 245W Protocol (Lindh and Smith 2024). TaqRed 2x Master Mix was used. The forward primer was GallWasp Forward 1 5'-ATTCAACCAATCATAAAGATATTGG -3', and the reverse primer was Gall Wasp Reverse 5'-TAAACTTCAGGGTGACCAAAAAATCA-3'. The thermocycler program was 94°C for one minute, 40 cycles of 94°C for 40 seconds, 45°C for 40 seconds, and 72°C for one minute, followed by 75°C for five minutes. PCR product was held at 4°C until use. Success of amplification was determined by gel electrophoresis. PCR products were cleaned using Exo-CIP-it, with a thermocycler program of 37°C for four minutes and 80°C for one minute. Samples were sent to Eurofins to be Sanger sequenced. Sequencing results were edited using CodonCode Aligner v12.0.4.

Genetic analysis

Sequences were assembled using the default settings in CodonCode, exported as interleaved Phylip files, and submitted to PhyML 3.0 on default settings to construct phylogenies using maximum likelihood (Guindon et al. 2010). Phylogenies were then analyzed and edited for visual clarity in FigTree v1.4.4 (Rambaut 2018). For closely related clades, a 2% branch length difference threshold was used to distinguish them (Stahlhut et al. 2013).

Data analysis

Once landmarked, wings were aligned using a generalized Procrustes alignment (`gpagen :: geomorph`, Adams et al. 2025). For each group of specimens analyzed, Principal Component Analysis (PCA) was then performed on the aligned landmarks (`gm.prcomp :: geomorph`). To visualize PCA results, `ggplot2` was used to plot coordinates according to the first two principal component axes (`ggplot :: ggplot2`, Wickham 2016). Analysis of Variance (ANOVA) was performed on the PCA results to test for significance, with the standard $p < 0.05$ as the threshold for significance (`procD.lm :: geomorph`). Canonical Variate Analysis (CVA) was performed with a Jackknife crossvalidation on the Procrustes aligned landmarks with grouping factors of sex, subgenus, and species/clade as shown in Table 1 (CVA :: Morpho, Schlager 2017).

All data analysis was performed and all figures unless otherwise specified were generated using R Studio version 2025.09.0.

Results

Phylogenetic

Samples with the same definitive BOLD classification grouped together in our phylogeny (Figure 1). With Roch's samples, our unidentified samples grouped into clades with a .02 branch length difference, which were called Clade X and Clade CY. Clade CY is composed of two clades, Clade C and Clade Y (Figure 1). Clade C contained almost all of the samples from Roch's Clade C and separated into a distinct clade even with the addition of Oregon sequences. Roch's Clades A and B interspersed throughout Clades X and Y with the addition of the Oregon sequences, and thus the clades observed here were given different names. Clade X and Clade C have a branch length difference of 0.024, but Clade X and Clade Y have a branch length difference of 0.015. Given this doesn't surpass the 0.02 branch length difference expected of Hymenoptera species, Clades C and Y were instead considered as one group, which, as stated above, was distinct from Clade X.

Most *Callimelissodes* sequences were sufficiently different from the rest of the sequences that CodonCode did not consider them similar enough to assemble together, and thus are not featured in the phylogeny.

Table 1 shows which samples were sorted into which clade, and their BOLD identification if applicable and their clade if BOLD identification was inconclusive or *M. agilis*.

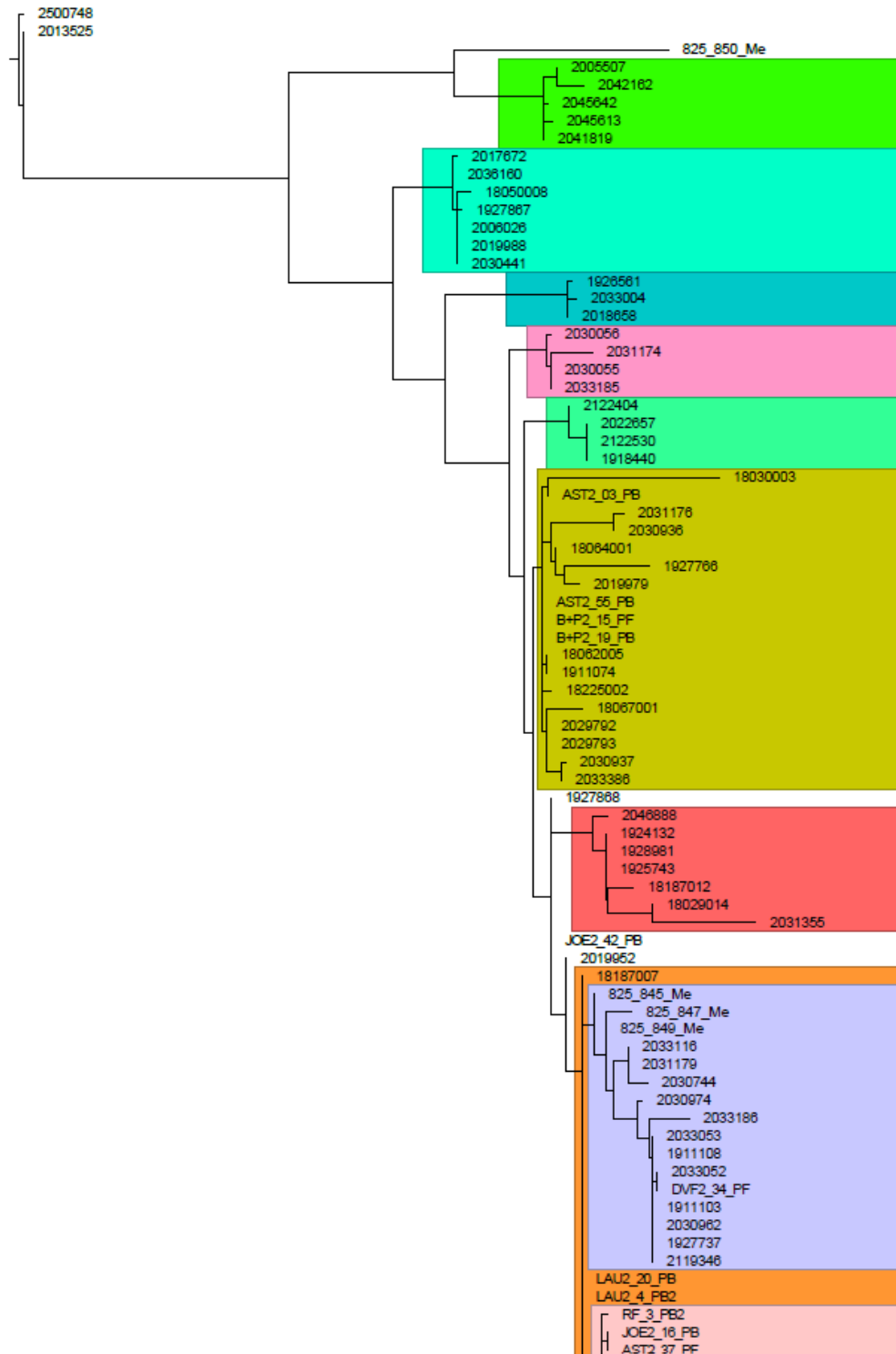


Figure 1 (continued on next page). Phylogeny with Oregon specimens and specimens from Roch 2023. Highlighted clades are, in descending order: *Mel. communis*, *Eu. robustior*, *Eu. saponella*, *Eu. utahensis*, *Helio. rivalis*, Clade X, *Eu. bimatrix*, and Clade CY, which contains Clade Y and Clade C.

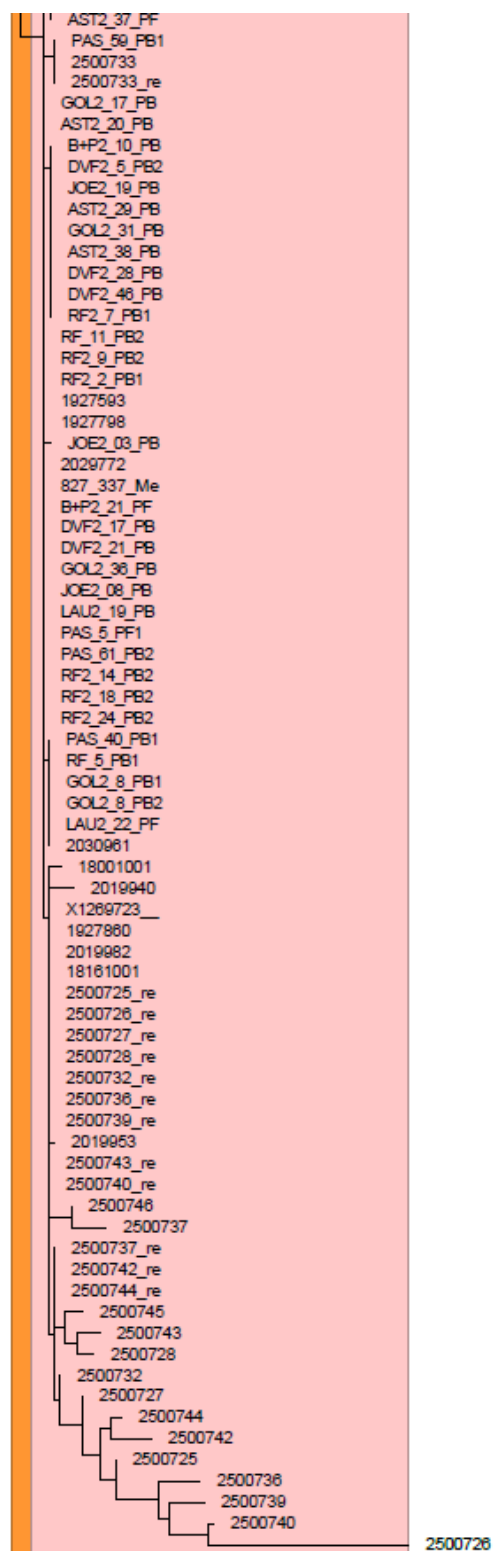


Figure 1 continued.

Morphometric

All subgenera

Principal Component Analysis (PCA) was first conducted on the entire dataset (Figure 2A), and then separated by sex to visualize the dimorphism between them (Figure 2B). PCA was then conducted on each sex separately (Figure 2C and D).

When Procrustes ANOVA was performed on the entire dataset, sex ($df=1,86$, $F=20.314$, $p = 0.001$), subgenus ($df = 3, 84$, $F = 3.581$, $p = 0.001$), and species/clade ($df = 10, 77$, $F = 3.706$, $p = 0.001$) were shown to have significant impacts on the placement of wing veins. When conducted on each sex separately, subgenus ($df = 4, 45$, $F = 2.402$, $p = 0.001$) and species/clade ($df = 10, 39$, $F = 2.673$, $p = 0.001$) have significant impacts on shape in female, while species/clade ($df = 7, 31$, $F = 2.496$, $p = 0.001$) has a significant impact in males, and subgenus has a slightly significant impact ($df = 1, 37$, $F = 2.314$, $p = 0.03$)

Canonical Variate Analysis (CVA) was then performed with sex, subgenus, and species/clade as grouping factors. Across the grouping factors, CVA was able to assign over 90% of the specimens to their predetermined group: sex ($K = 0.977$, 98.86% accuracy), subgenus ($K = 0.632$, 90.91% accuracy), and species/clade ($K = 0.939$, 95.45% accuracy). When conducted on each sex separately, the accuracy of assigning females to their correct subgenus decreased ($K = .382$, 69.39% accuracy), but the accuracy of assigning both females and males to their correct species/clade increased to 100% ($K = 1$).

Eumelissodes

As done with all subgenera, Principal Component Analysis (PCA) was conducted on the entire dataset (Figure 3A), and then separated by sex to visualize the dimorphism between them (Figure 3B). PCA was then conducted on each sex separately (Figure 3C and D).

When Procrustes ANOVA was performed on the combined dataset of both sexes in just *Eumelissodes*, sex ($df = 1, 73$, $F = 15.531$, $p = 0.001$) and species/clade ($df = 7, 67$, $F = 3.312$, $p = 0.001$) were shown to have significant impacts on the placement of wing veins. When conducted on each sex separately, species/clade was found to be significant in males ($df = 6, 30$, $F = 2.394$, $p = 0.001$) and females ($df = 6, 31$, $F = 2.435$, $p = 0.001$).

When CVA was conducted on the combined dataset of both sexes in just *Eumelissodes*, it was able to sort them by sex ($K = 0.920$, 96.00% accuracy) and by species/clade ($K = 0.919$, 94.67% accuracy). When conducted on each sex separately in just *Eumelissodes*, 100% ($K=1$) of specimens in both males and females were sorted into their predetermined species/clade.

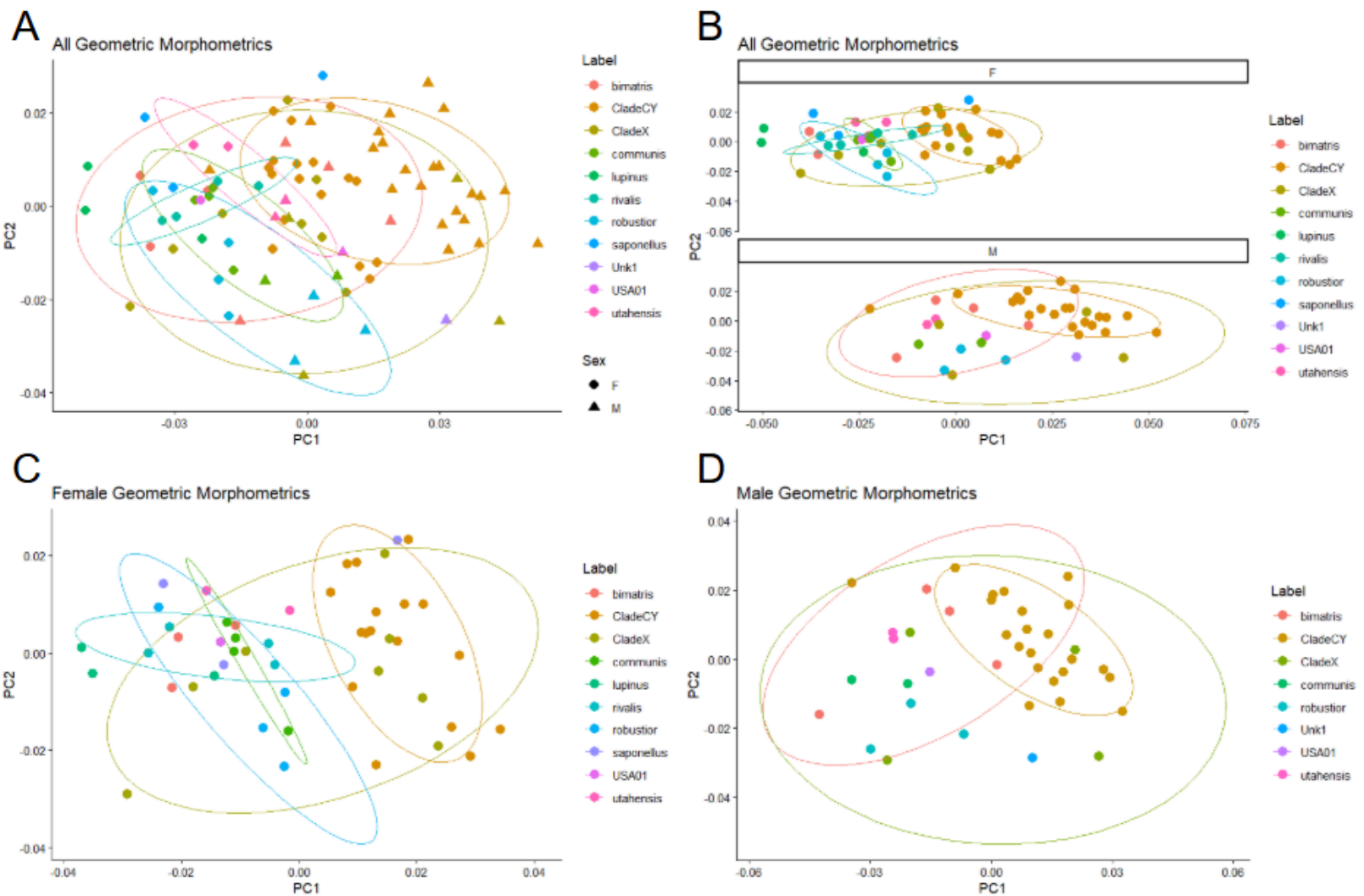


Figure 2. PCA of all subgenera. (A) PCA of all specimens, with species/clade distinguished by color and sex distinguished by shape. (B) PCA of all specimens separated by sex on the same principal component axes. (C) PCA of just female specimens. (D) PCA of just male specimens.

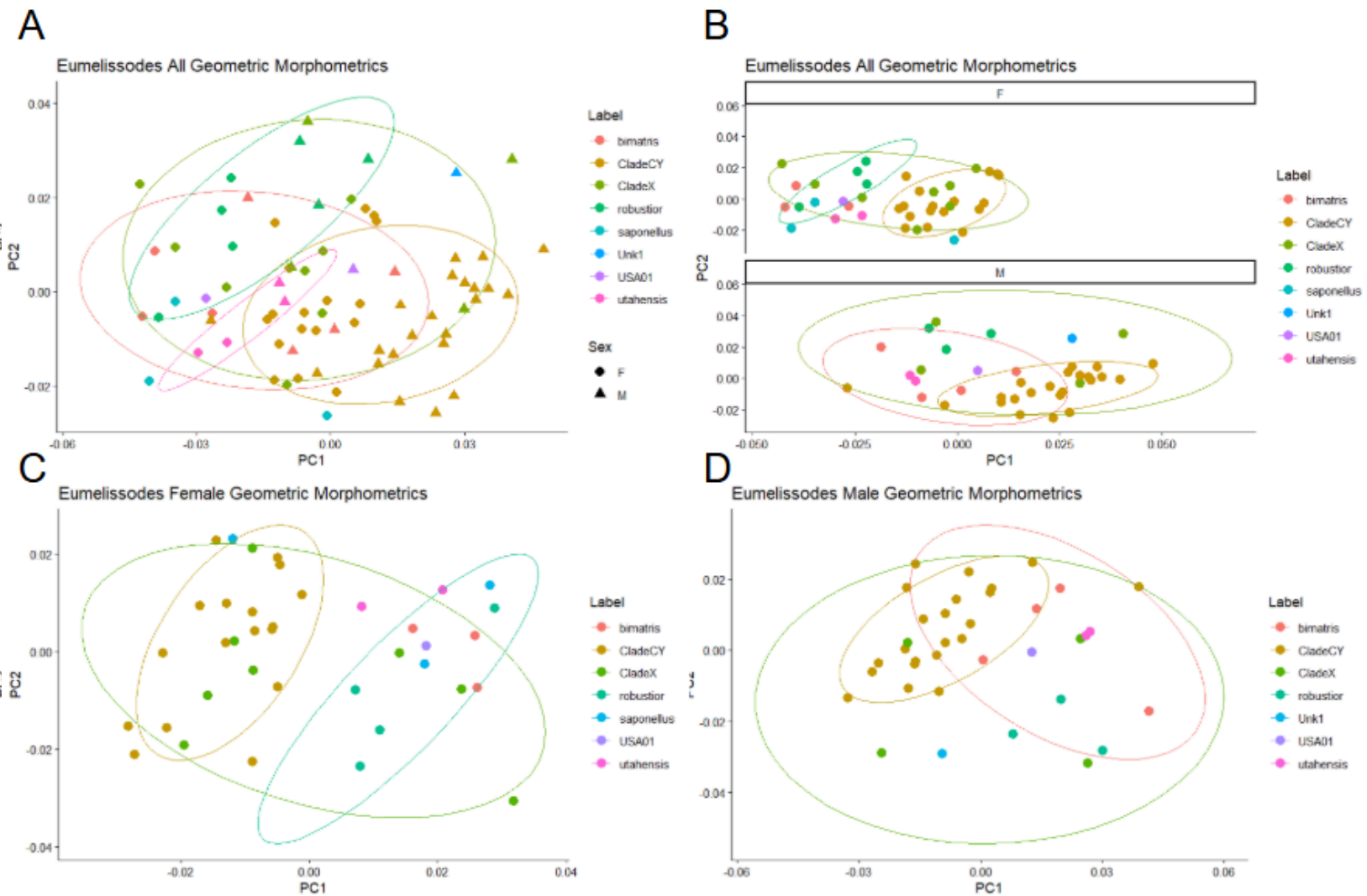


Figure 3. PCA of *Eumelissodes*. (A) PCA of all *Eumelissodes* specimens, with species/clade distinguished by color and sex distinguished by shape. (B) PCA of all *Eumelissodes* specimens separated by sex on the same principal component axes. (C) PCA of just female specimens. (D) PCA of just male specimens.

Discussion:

Phylogenetic analysis:

With the addition of the Willamette specimens, Roch's described clades A and B dispersed into Clades X and Y, while Roch's Clade C remained nearly intact in our phylogeny, with one specimen previously in C now in Y, and one previously in A now in C. However, Roch's subclades weren't visible with our additional sequences. Our results support Roch's claim that the current understanding of *M. agilis* and *M. trinodis* is inaccurate, but rather than point toward 3-5 different species, these results indicate that there may just be one species that contains both *M. agilis* and *M. trinodis*, and another unrealized species not closely related to *M. agilis* or *M. trinodis* (Roch 2023).

Morphometric analysis:

For being the clade with the highest number of specimens, Clade CY has a relatively tight and comprehensive ellipsis around the specimens. With the inclusion of non-*agilis* or -*trinodis* species, we can see the range of differences a well established *Melissodes* species has, and thus compare the differences between the patterns we see in the *M. agilis/trinodis* complex to what we expect a *Melissoodes* species to look like (Figures 2 and 3). This comparison is made difficult by the limited number of specimens from the other species represented here, but we do see that Clade CY has the range of differences similar to other species, but Clade X has differences beyond what we expect for a singular species. This mechanism for comparison would be greatly helped by an increased number of specimens in all species/clades.

The ANOVA on all subgenera confirms that the differences between clades and sexes visualized in the PCA depicted in Figure 2 are significant. Unsurprisingly, sex had both a low p-value and the highest F-value, which confirms the expected sexual

dimorphism in wing shape. Likewise, the separated sexes have significant p-values for both subgenus and species/clade, with the only group with a p-value greater than 0.001 being the subgenus separation in males, with a p-value of 0.03. This is likely due to the limited representation of males across subgenera, with males only appearing in *Eumelissodes* and *Melissodes*.

The CVA also confirmed these strong relationship, and the result of greatest note was the increased in accuracy of assigning species/clade when the sexes were separated. This, and the overlap between females and males we can see in Figure 2B, indicates that a female of one species may be more similar to a male of a different species than it is to a male of its own species. This, as well as the sexual dimorphism discussed earlier, suggests that for the most accurate morphometric results, specimens should be sexed and then analyzed in their separate groups.

We continued with *Eumelissodes* as the subgenus of interest, as that is where *M. agilis/trinodis* is, and there were simply not enough specimens of the other subgenera to make them worth analyzing separately. This also gives us slightly less cluttered PCA, but on the most part, the trends discussed above remain the same. Likewise, the ANOVA for both sex and species/clade in the combined sexes and each sex individually indicates the differences in shape remain significant.

Interestingly, in the CVA, the accuracy of assigning both sex and species/clade decreased by a few percentage points when looking at *Eumelissodes*, but when the sexes were separated, the accuracy of assigning species/clade went back up to 100%. This indicates, unsurprisingly, that *Eumelissodes* species have wing shapes more similar to each other than to other subgenera.

Conclusion:

Our results confirm the relationship between *Melissodes* species and wing shape, and lend more insight into the cryptic species complex that is *M. agilis* and *M. trinodis*. We confirm that there is likely an undescribed *Eumelissodes* species, and suggest that *M. agilis* and *M. trinodis* are likely the same species.

Citations

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Appendix

Table 1: Specimen accession numbers, sex, species/clade, and subgenus. Subgenera assignment via LaBerge 1956 and 1961.

Accession	Sex	Species/Clade	Subgenus
1911074	M	CladeX	Eumel
1911103	F	CladeCY	Eumel
1911108	F	CladeCY	Eumel
1918440	F	rivalis	Helio
1921399	F	communis	Meli
1924132	F	bimatrix	Eumel
1925743	F	bimatrix	Eumel
1926561	F	saponellus	Eumel
1927593	M	CladeCY	Eumel
1927737	F	CladeCY	Eumel
1927766	F	CladeX	Eumel
1927798	M	CladeCY	Eumel
1927860	M	CladeCY	Eumel
1927867	M	robustior	Eumel
1927868	M	Unk1	Eumel
1928981	F	bimatrix	Eumel
2003919	F	lupinus	Calli
2005507	F	communis	Meli
2006026	F	robustior	Eumel
2013525	F	lupinus	Calli
2017672	M	robustior	Eumel
2018658	F	saponellus	Eumel
2019940	F	CladeCY	Eumel
2019952	F	CladeCY	Eumel
2019953	F	CladeCY	Eumel
2019979	M	CladeX	Eumel
2019982	M	CladeCY	Eumel
2019988	M	robustior	Eumel
2022657	F	rivalis	Helio
2023468	F	lupinus	Calli
2029772	M	CladeCY	Eumel

2029792	F	CladeX	Eumel
2029793	F	CladeX	Eumel
2030055	F	utahensis	Eumel
2030056	M	utahensis	Eumel
2030441	F	robustior	Eumel
2030744	M	CladeCY	Eumel
2030936	F	USAo1	Eumel
2030937	M	CladeX	Eumel
2030961	M	CladeCY	Eumel
2030962	M	CladeCY	Eumel
2030974	F	CladeCY	Eumel
2031174	M	utahensis	Eumel
2031176	M	USAo1	Eumel
2031179	M	CladeCY	Eumel
2031355	M	bimatrix	Eumel
2033004	F	saponellus	Eumel
2033052	M	CladeCY	Eumel
2033053	M	CladeCY	Eumel
2033116	M	CladeCY	Eumel
2033185	F	utahensis	Eumel
2033186	M	CladeCY	Eumel
2033386	M	CladeX	Eumel
2036160	F	robustior	Eumel
2041819	M	communis	Meli
2042162	M	communis	Meli
2045613	F	communis	Meli
2045642	F	communis	Meli
2046888	M	bimatrix	Eumel
2119346	M	CladeCY	Eumel
2122404	F	rivalis	Helio
2122530	F	rivalis	Helio
2500725	M	CladeCY	Eumel
2500726	F	CladeCY	Eumel
2500727	F	CladeCY	Eumel
2500728	F	CladeCY	Eumel

2500732	M	CladeCY	Eumel
2500733	M	CladeCY	Eumel
2500736	F	CladeCY	Eumel
2500737	F	CladeCY	Eumel
2500739	F	CladeCY	Eumel
2500740	M	CladeCY	Eumel
2500742	F	CladeCY	Eumel
2500743	M	CladeCY	Eumel
2500744	M	CladeCY	Eumel
2500745	M	CladeCY	Eumel
2500746	M	CladeCY	Eumel
18001001	F	CladeCY	Eumel
18029014	M	bimatrix	Eumel
18030003	F	CladeX	Eumel
18050008	F	robustior	Eumel
18062005	F	CladeX	Eumel
18064001	F	CladeX	Eumel
18067001	F	CladeX	Eumel
18161001	F	CladeCY	Eumel
18187007	F	CladeCY	Eumel
18187012	M	bimatrix	Eumel
18225002	F	CladeX	Eumel